GRADE 7 ACCELERATED · UNIT 5 · PRACTICE TEST

# Multi-Step Proportional Relationships

*24 points* **WORKED ANSWERS**

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*Show your work. One point for each lettered part.*

## Part 1. Scale Drawings and Scale Factors *10 points*

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.*

**1.** A 4 cm side of a figure is 12 cm in its scale copy. Find the scale factor. *1 pt*

***k = 3***

*copy ÷ original*

*Scoring: 1 point.*

**2.** A 12 cm side of a figure is 6 cm in its scale copy. Find the scale factor. *1 pt*

***k = 0.5***

*copy ÷ original; the copy is smaller, so k is under 1 — 2 is the proportion backwards*

*Scoring: 1 point.*

**3.** A figure is scaled by a scale factor of 2. Select ALL of the statements that are true about the copy. *1 pt*

- ☐ A. The area is 2 times as large.
- ☒ **B. Each side is 2 times as long.**
- ☒ **C. The perimeter is 2 times as long.**
- ☒ **D. The area is 4 times as large.**
- ☐ E. Each angle is 2 times as large.

***B, C and D***

*All three and nothing else for the point: sides and perimeter × k, area × $k^2$, angles unchanged.*

*Scoring: 1 point.*

**4.** A 3 cm by 5 cm rectangle is scaled by k = 2.

**a.** Find the copy's perimeter. *1 pt*

***32 cm***

*perimeter × k*

**b.** Find the copy's area. *1 pt*

***60 cm²***

*area × k² — 30 multiplied the area by k once*

*Scoring: a 1 · b 1 = 2 points.*

**5.** A figure has area 6 cm². Its scale copy has area 150 cm². What is the scale factor? *1 pt*

***k = 5***

*150 ÷ 6 = 25 is k², so k = √25. A board with 25 handed in the area ratio as k.*

*Scoring: 1 point.*

**6.** A floor plan is drawn at 1 in : 5 ft. A room is 2 in by 3 in on the plan.

**a.** Find the real length of the wall drawn 2 in long. *1 pt*

***10 ft***

*the scale per 1 inch, then ×: 1 in : 5 ft*

**b.** Find the room's real area. *1 pt*

***150 ft²***

*real dimensions first: 10 by 15 — 6 in² × 5 = 30 is the named error*

*Scoring: a 1 · b 1 = 2 points.*

**7.** A plan is drawn at 1 in : 3 ft. A real wall is 15 ft long. How long is it on the plan? *1 pt*

***5 in***

*real ÷ rate*

*Scoring: 1 point.*

**8.** A plan is drawn at 1 in : 6 ft. A room is 1.5 in by 2.5 in on the plan. Find the room's real area. *1 pt*

***135 ft²***

*9 ft by 15 ft.*

*Scoring: 1 point.*

## Part 2. Converting Across Systems

*8 points*

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.*

**9.** 1 in = 2.54 cm. How many centimeters is 7 in? *1 pt*

***17.78 cm (17.9 by ÷ 0.39)***

*× the ratio whose 1 is inches; the sheet's other ratio, ÷ 0.39, earns the point too when the work shows it*

*Scoring: 1 point.*

**10.** 1 in = 2.54 cm and 1 cm = 0.39 in. Select ALL of the expressions that give the number of inches in 110 cm. *1 pt*

- ☐ A. 110 × 2.54
- ☒ **B. 110 × 0.39**
- ☐ C. 110 ÷ 0.39
- ☒ **D. 110 ÷ 2.54**

***B and D***

*Both sheet routes (42.9 and 43.3) and nothing else for the point.*

*Scoring: 1 point.*

**11.** 1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. How many centimeters is 1.5 yd? *1 pt*

***137.16 cm***

*a chain: 54 in, then × 2.54 (54 ÷ 0.39 = 138.5 by the other ratio — full credit). The estimate catches 3.81.*

*Scoring: 1 point.*

**12.** 1 kg = 2.204 lb. How many pounds is 35 kg? *1 pt*

***77.14 lb (77.09 by ÷ 0.454)***

*× the ratio whose 1 is the unit you have; the sheet's other ratio earns the point too when the work shows it*

*Scoring: 1 point.*

**13.** 1 gal = 3.785 L. How many liters is 3 gal? *1 pt*

***11.355 L (11.36 by ÷ 0.264)***

*more liters than gallons; the sheet's other ratio earns the point too when the work shows it*

*Scoring: 1 point.*

**14.** 1 in = 2.54 cm. A tile is 10 $in^2$ in area. Which is closest to its area in square centimeters? *1 pt*

- ○ A. 25.4 $cm^2$
- ● **B. 64.5 $cm^2$**
- ○ C. 6.45 $cm^2$
- ○ D. 50.8 $cm^2$

***B***

*$10 \times 2.54^2 = 64.52$ (or $10 \div 0.39^2 = 65.7$ — the sheet's two routes differ here; either picks B).*

*Scoring: 1 point.*

**15.** Rate on Nov 5: 1 US dollar = 1.32 Canadian dollars.

**a.** How many Canadian dollars is $90? *1 pt*

***118.80 Canadian dollars***

*× the rate*

**b.** How many US dollars is 66 Canadian dollars? *1 pt*

***$50***

*÷ the rate; 87.12 multiplied both ways*

*Scoring: a 1 · b 1 = 2 points.*

## Part 3. Multi-Step Proportion Problems *6 points*

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.*

**16.** A car gets 27 miles per gallon and its tank holds 10 gal.

**a.** How far does it go on a full tank? *1 pt*

***270 mi***

*the rate × the gallons*

**b.** It makes a 1,200-mi trip. How many tanks of gas does the trip use? (Part of a tank counts as a whole tank.) *1 pt*

***5 tanks (1,200 ÷ 270 ≈ 4.44)***

*the bracket on the paper settles the rounding; follow-through on part (a)*

*Scoring: a 1 · b 1 = 2 points.*

**17.** A recipe for 5 servings uses 2 cups of rice.

**a.** How much rice for 15 servings? *1 pt*

***6 cups***

*k = 3 from the yields, then the ingredient*

**b.** With 10 cups on hand, how many servings can be made? *1 pt*

***25 servings***

*5 recipes × the yield; 50 scaled the yield by the cups*

*Scoring: a 1 · b 1 = 2 points.*

**18.** Paint is mixed 3 : 1 : 2. A batch uses 4 gal of the second color. How many gallons is the whole batch? *1 pt*

***24 gal***

*k = 4 from the second color, then the whole, 6 parts; 12 found the first color and stopped*

*Scoring: 1 point.*

**19.** 1 mi = 1.61 km. A train does 80 km/h; a bus does 45 mi/h. Which is faster? Show both speeds in one unit. *1 pt*

***The train: 45 mi/h = 72.45 km/h, and 80 > 72.45.***

*Same unit first; 80 > 45 compared two units.*

*Scoring: 1 point.*